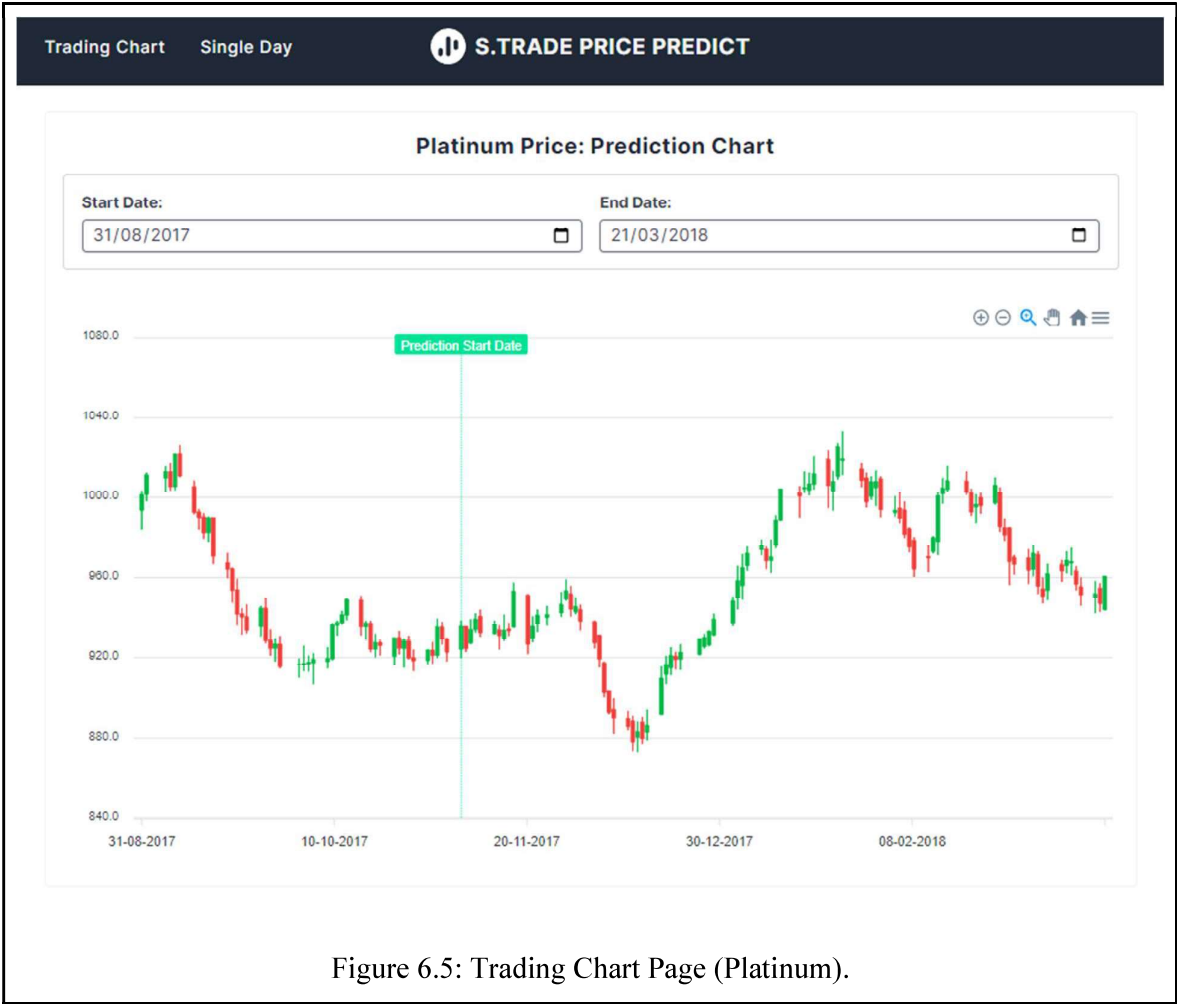




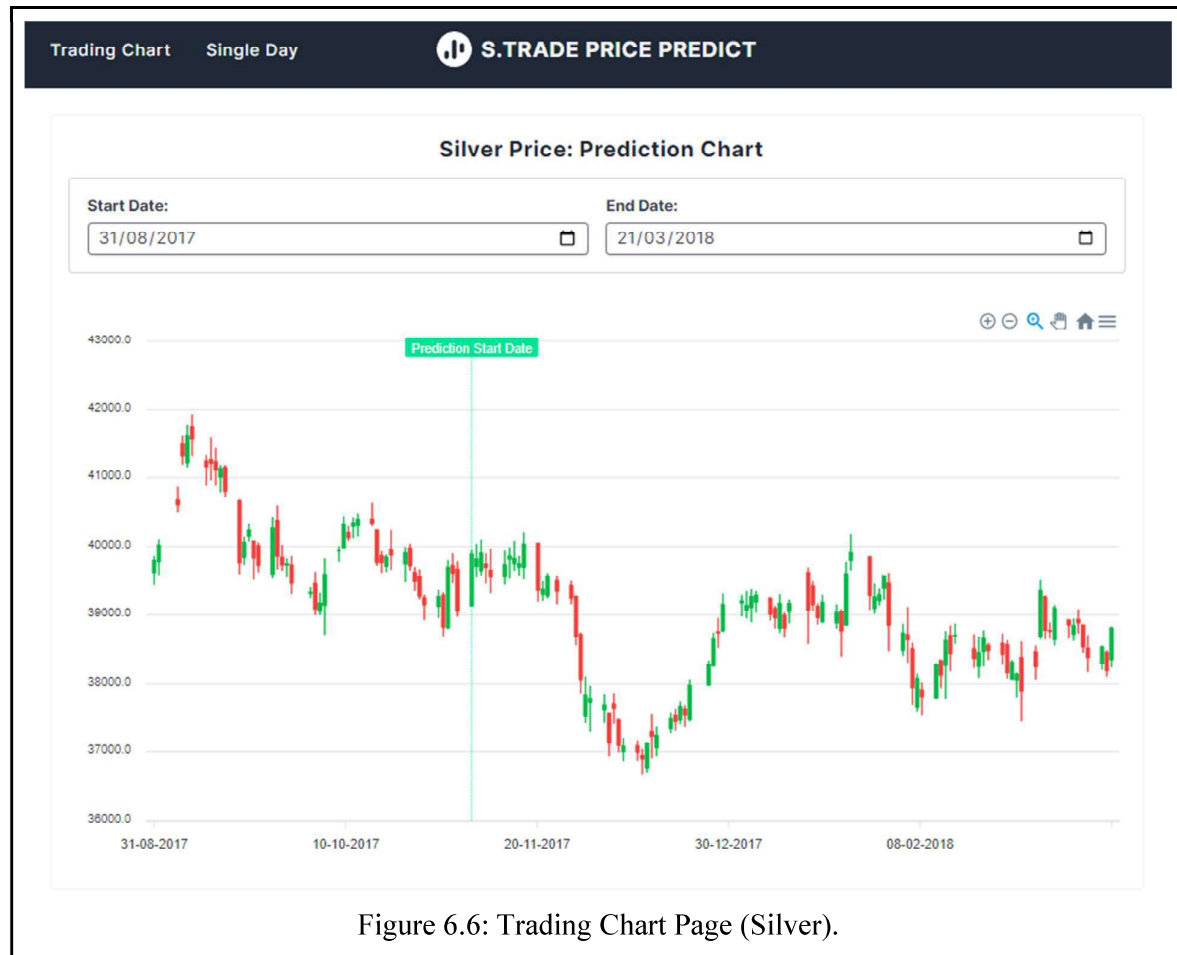


Figure 6.6: Trading Chart Page (Silver).

Figure 6.4, Figure 6.5 and Figure 6.7 shows the Trading Chart page of Gold, Platinum and Silver. The purpose of this trading chart page is to allow users to predict prices starting from the 'Prediction Start Date'. Take Figure 6.6 as an example, there is a long green vertical line in the chart stating "Prediction Start Date". This long green vertical line means that the price after this green line is the predicted price. It is based on the test data set, in which the prediction start date is the first instance of the test dataset for Gold, Platinum, and Silver respectively. By looking at the long green vertical line, users can know where the predicted price starts at. Adjusting the 'Start Date' and 'End Date' at the top can change the date range of the trading chart (candlestick chart).

Trading Chart

Single Day

 S.TRADE PRICE PREDICT

Gold Price: Single Day Prediction

GOLD HIGH Enter GOLD HIGH	GOLD LOW Enter GOLD LOW	GOLD OPEN Enter GOLD OPEN	GDX LOW Enter GDX LOW
GDX CLOSE Enter GDX CLOSE	GDX HIGH Enter GDX HIGH	GDX AJCLOSE Enter GDX AJCLOSE	GDX OPEN Enter GDX OPEN
SF LOW Enter SF LOW	SF PRICE Enter SF PRICE	SF OPEN Enter SF OPEN	SF HIGH Enter SF HIGH
EG LOW Enter EG LOW	EG OPEN Enter EG OPEN	EG CLOSE Enter EG CLOSE	EG HIGH Enter EG HIGH
EG AJCLOSE Enter EG AJCLOSE	PLT PRICE Enter PLT PRICE	PLT HIGH Enter PLT HIGH	PLT LOW Enter PLT LOW
PLT OPEN Enter PLT OPEN	OF HIGH Enter OF HIGH	OF PRICE Enter OF PRICE	OF OPEN Enter OF OPEN
OF LOW Enter OF LOW	SF VOLUME Enter SF VOLUME		

(*Optional) Auto Input Data from Existing Date:

dd/mm/yyyy



Set

Predict

Reset

Figure 6.7: Single Day Prediction Page (Gold).

Trading Chart

Single Day

 S. TRADE PRICE PREDICT

Platinum Price: Single Day Prediction

PLT HIGH Enter PLT HIGH	PLT LOW Enter PLT LOW	PLT OPEN Enter PLT OPEN	USO HIGH Enter USO HIGH
USO AJCLOSE Enter USO AJCLOSE	USO CLOSE Enter USO CLOSE	USO OPEN Enter USO OPEN	USO LOW Enter USO LOW
EG AJCLOSE Enter EG AJCLOSE	EG LOW Enter EG LOW	EG CLOSE Enter EG CLOSE	EG OPEN Enter EG OPEN
EG HIGH Enter EG HIGH	OF HIGH Enter OF HIGH	OF PRICE Enter OF PRICE	OF OPEN Enter OF OPEN
OF LOW Enter OF LOW	OS HIGH Enter OS HIGH	OS OPEN Enter OS OPEN	OS PRICE Enter OS PRICE
OS LOW Enter OS LOW	EU HIGH Enter EU HIGH	EU PRICE Enter EU PRICE	EU OPEN Enter EU OPEN
EU LOW Enter EU LOW	SF HIGH Enter SF HIGH	SF PRICE Enter SF PRICE	SF OPEN Enter SF OPEN
SF LOW Enter SF LOW	GOLD HIGH Enter GOLD HIGH	GOLD AJCLOSE Enter GOLD AJCLOSE	GOLD CLOSE Enter GOLD CLOSE
GOLD OPEN Enter GOLD OPEN	GOLD LOW Enter GOLD LOW	GDX HIGH Enter GDX HIGH	GDX OPEN Enter GDX OPEN
GDX CLOSE Enter GDX CLOSE	GDX LOW Enter GDX LOW	GDX AJCLOSE Enter GDX AJCLOSE	

(*Optional) Auto Input Data from Existing Date:



Set

Predict

Reset

Figure 6.8: Single Day Prediction Page (Platinum).

Trading Chart
Single Day

Silver Price: Single Day Prediction

SF HIGH Enter SF HIGH	SF LOW Enter SF LOW	SF OPEN Enter SF OPEN	GOLD HIGH Enter GOLD HIGH
GOLD CLOSE Enter GOLD CLOSE	GOLD AJCLOSE Enter GOLD AJCLOSE	GOLD LOW Enter GOLD LOW	GOLD OPEN Enter GOLD OPEN
GDX HIGH Enter GDX HIGH	GDX OPEN Enter GDX OPEN	GDX CLOSE Enter GDX CLOSE	GDX LOW Enter GDX LOW
GDX AJCLOSE Enter GDX AJCLOSE	EG LOW Enter EG LOW	EG OPEN Enter EG OPEN	EG CLOSE Enter EG CLOSE
EG HIGH Enter EG HIGH	EG AJCLOSE Enter EG AJCLOSE	PLT PRICE Enter PLT PRICE	PLT HIGH Enter PLT HIGH
PLT LOW Enter PLT LOW	PLT OPEN Enter PLT OPEN	OF HIGH Enter OF HIGH	OF PRICE Enter OF PRICE
OF OPEN Enter OF OPEN	OF LOW Enter OF LOW		

(*Optional) Auto Input Data from Existing Date:

☐

Figure 6.7, Figure 6.8 and Figure 6.9 show the ‘Single Day Prediction’ page for each precious metals (Gold, Platinum and Silver). ‘Single Day Prediction’ page is only used to predict price for a certain single day only. In order to predict the price, all the input fields must be filled in. Since there are too many input fields which might be time-consuming for users to fill in one-by-one, a user-friendly feature that allows automatic input to be filled into each field is provided. Users will just need to click onto the ‘dd/mm/yyyy’ input field, choose an available date (based on the dataset) and click the ‘Set’ button. After that, all the data that is linked to the selected date will be automatically filled into each input field. Once the ‘Predict’ button is clicked, the predicted price will be displayed.

Trading Chart

Single Day

 S. TRADE PRICE PREDICT

Silver Price: Single Day Prediction

SF HIGH	SF LOW	SF OPEN	GOLD HIGH
Enter SF HIGH	Enter SF LOW	Enter SF OPEN	Enter GOLD HIGH
GOLD CLOSE	GOLD AJCLOSE	GOLD LOW	GOLD OPEN
Enter GOLD CLOSE	Enter GOLD AJCLOSE	Enter GOLD LOW	Enter GOLD OPEN
GDX HIGH	GDX OPEN	GDX CLOSE	GDX LOW
Enter GDX HIGH	Enter GDX OPEN	Enter GDX CLOSE	Enter GDX LOW
GDX AJCLOSE	EG LOW	EG OPEN	EG CLOSE
Enter GDX AJCLOSE	Enter EG LOW	Enter EG OPEN	Enter EG CLOSE
EG HIGH	EG AJCLOSE	PLT PRICE	PLT HIGH
Enter EG HIGH	Enter EG AJCLOSE	Enter PLT PRICE	Enter PLT HIGH
PLT LOW	PLT OPEN	OF HIGH	OF PRICE
Enter PLT LOW	Enter PLT OPEN	Enter OF HIGH	Enter OF PRICE
OF OPEN	OF LOW		
Enter OF OPEN	Enter OF LOW		

(*Optional) Auto Input Data from Existing Date:

17/08/2017



Set

August, 2017

↑

↓

S	M	T	W	T	F	S
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9

Clear

Today

Figure 6.10: Selecting an Available Date.

By clicking on the date input field, the user can choose a date.

Trading Chart
Single Day

S. TRADE PRICE PREDICT

☒ Input complete!

Silver Price: Single Day Prediction

SF HIGH 39350	SF LOW 38957	SF OPEN 39000	GOLD HIGH 122.610001
GOLD CLOSE 122.489998	GOLD AJCLOSE 122.489998	GOLD LOW 121.760002	GOLD OPEN 122.290001
GDX HIGH 23.26	GDX OPEN 23.16	GDX CLOSE 23.030001	GDX LOW 22.950001
GDX AJCLOSE 22.725435	EG LOW 9.25	EG OPEN 9.65	EG CLOSE 9.35
EG HIGH 9.7	EG AJCLOSE 9.35	PLT PRICE 980	PLT HIGH 985.2
PLT LOW 974.55	PLT OPEN 982.45	OF HIGH 51.12	OF PRICE 51.03
OF OPEN 50.4	OF LOW 50		

(*Optional) Auto Input Data from Existing Date:

Figure 6.11: Automatically Input Data.

Once the date is chosen and the “Set” button is clicked, the data will be automatically input into all the input fields. Furthermore, by clicking on the ‘Reset’ button, all the input fields will be cleared.

Trading Chart
Single Day

S. TRADE PRICE PREDICT


Price predicted!

Silver Price: Single Day Prediction

SF HIGH 39350	SF LOW 38957	SF OPEN 39000	GOLD HIGH 122.610001
GOLD CLOSE 122.489998	GOLD AJCLOSE 122.489998	GOLD LOW 121.760002	GOLD OPEN 122.290001
GDX HIGH 23.26	GDX OPEN 23.16	GDX CLOSE 23.030001	GDX LOW 22.950001
GDX AJCLOSE 22.725435	EG LOW 9.25	EG OPEN 9.65	EG CLOSE 9.35
EG HIGH 9.7	EG AJCLOSE 9.35	PLT PRICE 980	PLT HIGH 985.2
PLT LOW 974.55	PLT OPEN 982.45	OF HIGH 51.12	OF PRICE 51.03
OF OPEN 50.4	OF LOW 50		

(*Optional) Auto Input Data from Existing Date:


Set

Predict

Reset


Predicted SILVER price : \$39256.649717

Figure 6.12: Automatically Input Data.

Once all the input fields already have data and the ‘Predict’ button is clicked, the predicted price will be displayed.

Trading Chart

Single Day

 S. TRADE PRICE PREDICT

This day has no trading data

Information:
1. Exist Dates: 15/12/2012 - 31/12/2018)
2. No trading on holidays

Silver Price: Single Day Prediction

SF HIGH

Enter SF HIGH

SF LOW

Enter SF LOW

SF OPEN

Enter SF OPEN

GOLD HIGH

Enter GOLD HIGH

GOLD CLOSE

Enter GOLD CLOSE

GOLD AJCLOSE

Enter GOLD AJCLOSE

GOLD LOW

Enter GOLD LOW

GOLD OPEN

Enter GOLD OPEN

GDX HIGH

Enter GDX HIGH

GDX OPEN

Enter GDX OPEN

GDX CLOSE

Enter GDX CLOSE

GDX LOW

Enter GDX LOW

GDX AJCLOSE

Enter GDX AJCLOSE

EG LOW

Enter EG LOW

EG OPEN

Enter EG OPEN

EG CLOSE

Enter EG CLOSE

EG HIGH

Enter EG HIGH

EG AJCLOSE

Enter EG AJCLOSE

PLT PRICE

Enter PLT PRICE

PLT HIGH

Enter PLT HIGH

PLT LOW

Enter PLT LOW

PLT OPEN

Enter PLT OPEN

OF HIGH

Enter OF HIGH

OF PRICE

Enter OF PRICE

OF OPEN

Enter OF OPEN

OF LOW

Enter OF LOW

(*Optional) Auto Input Data from Existing Date:

dd/mm/yyyy

Set

Predict

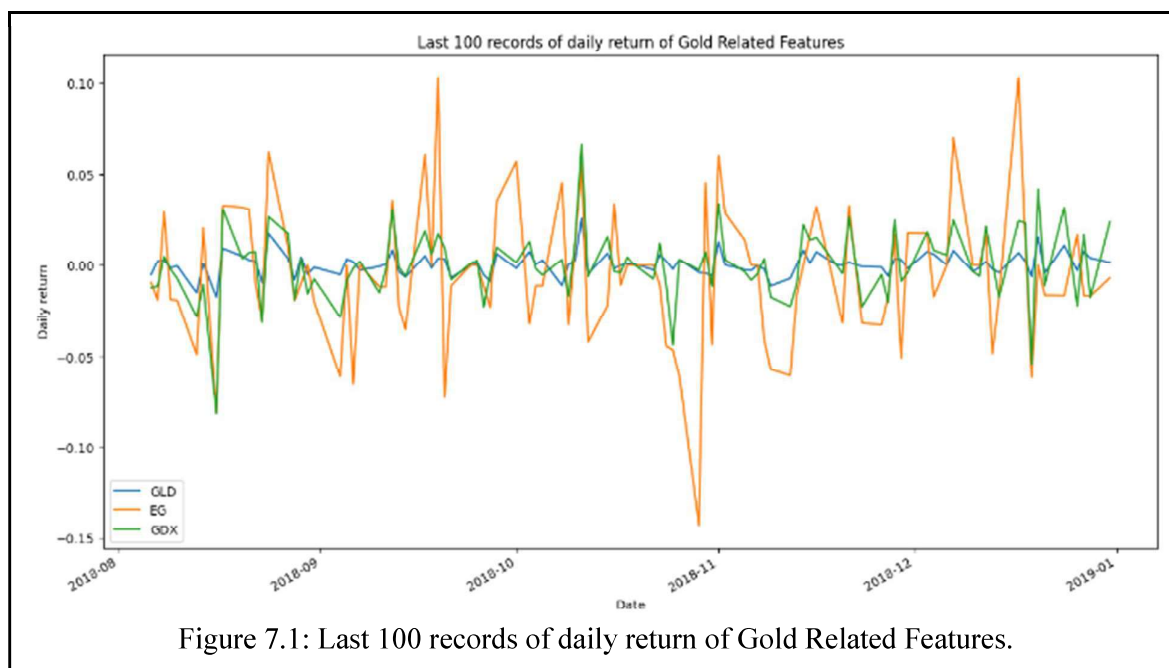
Reset

Figure 6.13: Invalid Date.

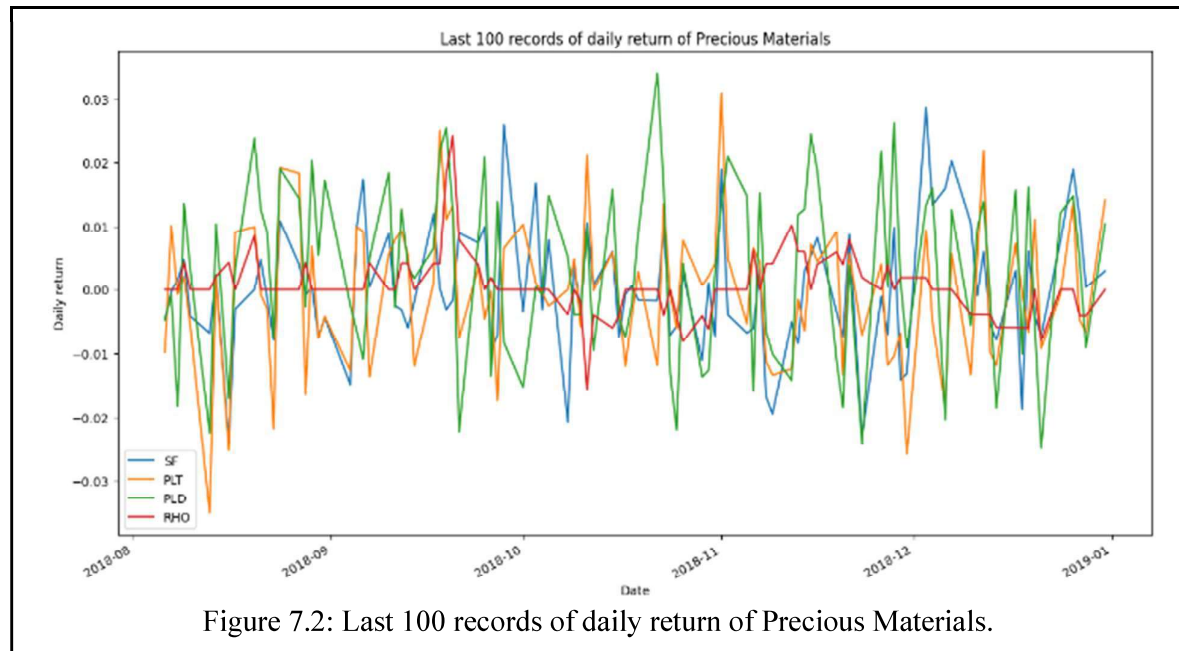
If the selected date has no trading data (the date is out of range or it is a Public Holiday/Saturday/Sunday), an error message will be displayed at the top right corner of the screen.

7.0 FURTHER ANALYSIS

Daily return is calculated and generated into a graph. Daily return represents the percentage change in the value of a financial asset, like precious metals, stocks or commodities during a single trading day. This metric is crucial for traders or trading companies like S.Trade as it allows them to gauge an asset's performance and volatility (Calzon, 2023). The daily return is calculated by comparing the closing price to the opening price (Kenton, 2021), daily returns can be positive, negative, or zero, signifying gains, losses, or no change. Hence, S.Trade can use the information from daily returns graph to gauge an asset's performance and volatility. Assets with high volatility present both opportunities and risks, while low-volatility assets offer stability with potentially lower returns. Not only that, S.Trade can use daily return data to develop and test trading strategies. They look for patterns or trends in the data that can be leveraged for profit, such as identifying buy or sell signals based on historical daily returns.



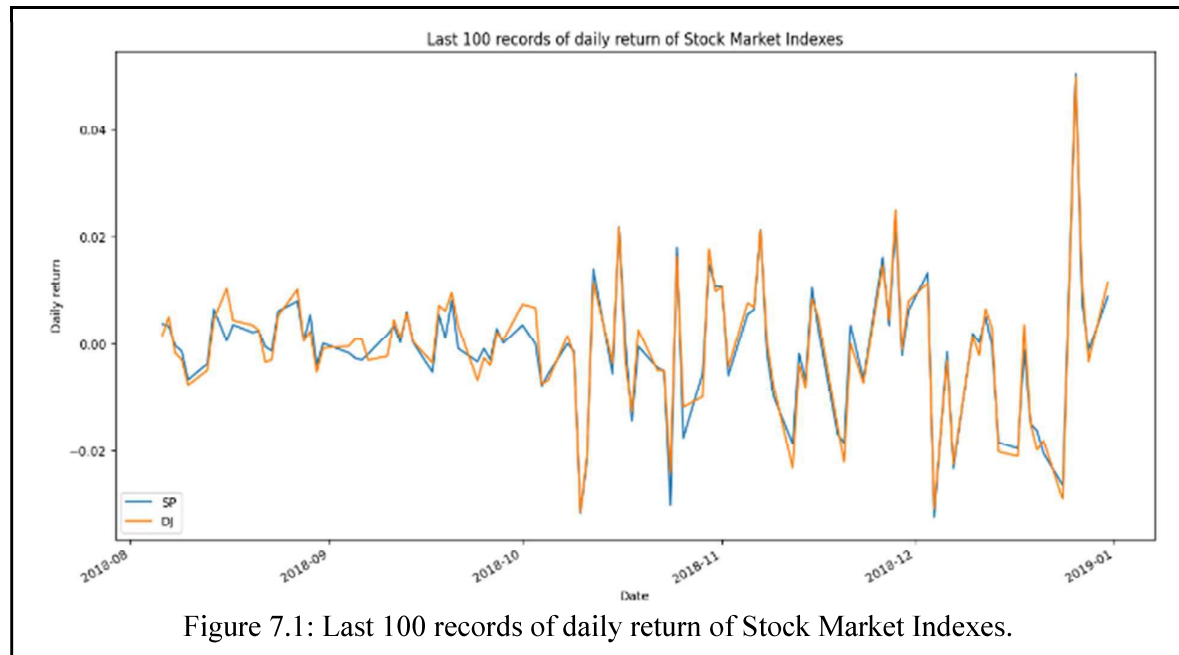
This graph shows the last 100 records of daily returns of Gold Related Features such as Gold(GLD), Eldorado Gold Corporation (EGO) and Gold Miners ETF(GDX).



This graph shows the last 100 records of daily returns of Precious Materials such as Silver(SF), Platinum(PLT), Palladium(PLD) and Rhodium(RHO). The reason why the graphs of these 4 Precious Metals are not aligned together is because according to (Economics, 2016), they are not highly correlated to each other. Refer to Table 7.1 below:

Table 7.1 Correlation Data Between 4 Precious Metals

Correlated Data				
	Platinum	Palladium	Silver	Rhodium
Platinum	1.00	0.60	0.74	0.20
Palladium	0.60	1.00	0.51	0.11
Silver	0.74	0.51	1.00	0.18
Rhodium	0.20	0.11	0.18	1.00



This graph shows the last 100 records of daily returns of Stock Market Indexes such as S&P 500 Index(SP) and Dow Jones Index(DJ). The S & P and Dow Jones Index moves almost the same and this is because the S & P and Dow Jones Index have a strong correlation with a value of 0.914 (Potharla, 2012). Hence, if S&P goes up, the Dow Jones Index will also go up. The same goes the other way round, if S&P goes down, the Dow Jones Index will also go down. With that being said, the daily return will be almost the same between S&P and Dow Jones Index.

